TFE4120 Exams June 2024

Total 100 marks

Exercise 1: 10 marks (2 and 8)

a) What is meant by materials' electric susceptibility (χ_e), and how is this related to polarization (P) for linear dielectrics?
b) Derive the relative permittivity (ε_r) using electric displacement (D) and P. How it relates to χ_e?

Solution:

a) The students should answer something like: susceptibility is the ability of materials to be polarized under an applied E-field. For linear dielectrics: $P = \varepsilon_o \chi_e E$.
b) $D = \varepsilon_o E + P = \varepsilon_o E + \varepsilon_o \chi_e E = (1 + \chi_e)\varepsilon_o E$, where $\varepsilon = (1 + \chi_e)\varepsilon_o$ is the material permittivity and therefore $D = \varepsilon E$. We define ε_r as:
$$\varepsilon_r = \frac{\varepsilon}{\varepsilon_o} = (1 + \chi_e)$$

Exercise 2: 10 marks (2, 2, 3 and 3)

a) Define the basic 3 categories of magnetic materials and how they respond to external magnetic field?
b) Considering linear magnetic media, define how magnetic susceptibility (χ_m) relates to magnetization (M)?
c) Derive the relative permeability (μ_r) using the Auxiliary field (H).
d) What is a nonlinear magnetic material? Do you know a category of those? What is a hysteresis loop (draw it as M vs I or B vs $\mu_o H$ and explain its basic points)?

Solution:

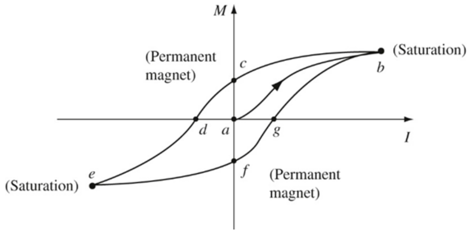
a) Paramagnetic with magnetization in the direction and parallel to the external magnetic field. Diamagnetic with magnetization in the opposite direction to the external magnetic field. Ferromagnets that preserve the magnetization even when the external magnetic field is removed.

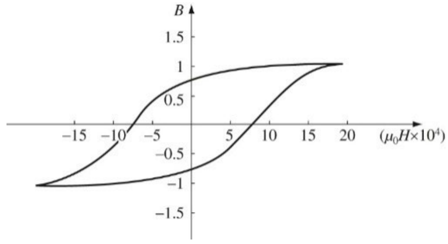
b) $M = \chi_m H$, where $H = (1/\mu_o)B - M$ is the Auxiliary field.

c) $H = \left(\frac{1}{\mu_o}\right) B - M \Rightarrow B = \mu_o(H + M) = \mu_o(H + \chi_m H) = \mu_o(1 + \chi_m)H = \mu H$, where μ is the permeability of the material. Further

$\mu = \mu_o(1 + \chi_m)$ and $\mu_r = \frac{\mu}{\mu_o} = (1 + \chi_m)$ is the relative permeability.

d) A nonlinear magnetic material is not obeying the $M = \chi_m H$, meaning that more H does not result necessary to more M. A category of nonlinear magnetic materials are the ferromagnets. The students should draw one of the following:





Exercise 3: 20 marks (10 and 10)

Prove the following Maxwell's equations:

a) $\nabla \times B = \mu_o J$ considering Ampere's Law, think how to relate total current enclosed to the volume current density J and then apply Stoke's theorem.

b) $\nabla \cdot E = \frac{p}{\varepsilon_o}$, considering Gauss's Law and divergence theorem then think how to relate the total charge enclosed to the charge density p.

Solution:

a) For $\nabla \times B = \mu_o J$ proof briefly:

Ampere Law in integral form is given as $\oint B \cdot dI = \mu_o I_{enc}$,
Where the enclose current is $I_{enc} = \int J \cdot da$

Applying the Stokes' theorem to the close integral $\oint B \cdot dI = \mu_o I_{enc} = \mu_o \int J \cdot da$

$$\int \nabla \times B \cdot da = \mu_o \int J \cdot da$$

Dropping the same integrals from both sides results in:

$$\nabla \times B = \mu_o J$$

b) For $\nabla \cdot E = p/\varepsilon_o$ proof briefly:

Gauss's Law in the integral form is given as $\oint E \cdot da = \frac{Q_{enc}}{\varepsilon_o}$.

Applying the divergence theorem for E relates a closed surface to a volume integral:

$$\oint E \cdot da = \int (\nabla \cdot E) d\tau$$

Now Q_{enc} is relates to volume integral as follows:

$$Q_{enc} = \int p \, d\tau$$

Then the above divergence theorem becomes:

$$\int (\nabla \cdot E) d\tau = \int p/\varepsilon_o \, d\tau$$

Dropping the volume integrals from both sides results in:

$$\nabla \cdot E = p/\varepsilon_o$$

Exercise 4: 20 Marks (10 and 10)

a) We have a 'thick' wire that has circular cross-section of radius: a. The wire carries a volume current density $J = ks$, where k is constant. What is the total current of the wire?
b) The same wire now carries a uniform current $I = 3i^2$. What is the J?

Hint: Think what formulas we use for uniform current densities (constant charge flow). In 1), J is varying as a function of the radius. What does that mean? Use cylindrical coordinates.

Solution:

a) Here the current is varying thus we need to use surface integral of $I = \int J \cdot da$.

$$I = \int J \cdot da = \int_0^{2\pi} \int_0^a ks\,(sds\,d\varphi) = 2\pi k \int_0^a s^2\,ds = \frac{2\pi k a^3}{3}$$

b) Now we have a uniform current: $I = 3i^2$. We find the surface area of the wire: $\pi(a)^2$
The $J = \frac{dI}{da}$, da perpendicular to the flow. $J = \frac{3i^2}{\pi(a)^2}$.

Exercise 5: 20 Marks (10 and 10)

a) How does the Electric Potential (V) relate to E-field.

b) Calculate V inside and outside of a spherical shell with radius r. The shell has a charge of $2q$.

Solution:

a) We expect the students to answer like this: $V = -\int_o^r E \cdot dl$, o a reference point, or in differential form: $E = -\nabla V$, and short explanation.

b)

Outside the shell: design a Gaussian surface R_{Gauss}>r.

Therefore Q_{enc}=$2q$,

$$\oint E \cdot da = \frac{Q_{enc}}{\varepsilon_o}$$

$$\oint E \cdot da = E(4\pi R_{Gauss}^2)\hat{r} = \frac{2q}{\varepsilon_o} => E = \frac{q}{2\pi\varepsilon_o R_{Gauss}^2}\hat{r}.$$

Inside the shell: design a Gaussian surface R_{Gauss}<r.

Here Q_{enc}=0 as it is a shell, E=0.

V outside: Here the students should consider integrating from infinity to the Gaussian surface, meaning:

$$V = -\int_o^r E \cdot dl = -\int_\infty^{R_{Gauss}} \frac{1}{2\pi\varepsilon_o}\frac{q}{r^2}\,dr = \frac{1}{2\pi\varepsilon_o}\frac{q}{R_{Gauss}}$$

V inside: Here the students should consider breaking the integral from infinity to the surface of the shell and from the shell to the Gaussian surface, meaning:

$$V = -\int_o^r E \cdot dl = -\int_\infty^r \frac{1}{2\pi\varepsilon_o}\frac{q}{r^2}dr - \int_r^{R_{Gauss}} 0dr = \frac{1}{2\pi\varepsilon_o}\frac{q}{r}.$$

Overall, the fundamental difference between E-field and V is that V is not zero inside the shell, even though the E-field is vanishing!

Exercise 6: 20 Marks (10 (2, 4 and 4) and 10 (5 and 5))

A conducting sphere of radius R contains a uniform charge of $tQ^2/2$. It is surrounded by a thick rubber of radius a with permittivity ε.

a) Find the E-field...
   i. in the sphere
   ii. in the rubber
   iii. out (i.e. outside the rubber/sphere system)

Think what you can find first considering what is given above.

b)
   i. Considering linear dielectrics, find the polarization (P) in the system (assume χ_e is known). Think where P should be located?
   ii. After answering this, find the volume (p_b) and surface (σ_b) bound charges. Use spherical coordinates. Think how many surfaces the insulating material has?

Solution:

a) The students should think that charge ($tQ^2/2$) is producing 'free' charges. Then using Gauss's law for the electric displacement (D):

In general $r > R$:

$$\oint D \cdot da = Q_{f_{enclosed}} => D(4\pi r^2) = tQ^2/2 => D = \frac{tQ^2}{8\pi r^2}\hat{r}$$

i. Now the E-field for $r < R$:

$E = D = P = 0$ as this is a conductor.

ii. For $a > r > R$:

We know (Ex. 1b) that $D = \varepsilon_o E + P = \varepsilon_o E + \varepsilon_o \chi_e E = (1 + \chi_e)\varepsilon_o E$, where $\varepsilon = (1 + \chi_e)\varepsilon_o$ is the material permittivity and therefore $D = \varepsilon E$.

$$E = \frac{tQ^2}{8\pi\varepsilon r^2}\hat{r}$$

iii. For $r > a$: $D = E\varepsilon_o + P$, $P = 0$ as this is empty space, no dielectric medium there. Therefore:

$$E = D/\varepsilon_o = \frac{tQ^2}{8\pi\varepsilon_o r^2}\hat{r}$$

b)

i.

Now that we know E-field everywhere we can calculate the polarization and the bound charges produced by the dielectric material due to the imposed charge of the metallic sphere.

$$P = \varepsilon_o \chi_e E = \frac{\varepsilon_o \chi_e t Q^2}{8\pi\varepsilon r^2}\hat{r}$$

ii.

After finding polarization, we can now find bound charges, starting from p_b in spherical coordinates:

$$p_b = -\nabla \cdot P = -\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\varepsilon_o \chi_e t Q^2}{8\pi\varepsilon r^2}\right) = 0$$

For $\sigma_b = P \cdot \hat{n}$, you have two surfaces of the insulating material: one inner that $\hat{n}$ shows towards the center of the sphere (therefore negative sign) and an outer where $\hat{n}$ shows out radially (positive sign).

Outer: $\sigma_b = P \cdot \hat{n} = \frac{\varepsilon_o \chi_e t Q^2}{8\pi\varepsilon a^2}$

Inner: $\sigma_b = P \cdot \hat{n} = -\frac{\varepsilon_o \chi_e t Q^2}{8\pi\varepsilon R^2}$.